

Tianyi ZHANG

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Objective

Seeking semiconductor engineering opportunities at AMD Shanghai, with a focus on SoC design and verification, automation infrastructure, AI-assisted EDA tooling, and hardware/software workflow optimisation.

Education

MSc Precision Scientific Instrumentation Expected 2026 – 2027
Nanyang Technological University Singapore

- Focus on semiconductor instrumentation, precision electronics, sensing systems, advanced hardware integration, and artificial intelligence.

BEng Mechanical Engineering (First Class Honours) Sept 2023 – Jun 2026
University College London London, United Kingdom

- First Class Honours, approximately equivalent to GPA 3.7+/4.0 in the Chinese grading system.
- Relevant coursework: control systems, computational modelling, algorithm development, embedded systems, and real-time systems.

Skills

Programming: Python, Java, C++, MATLAB, Shell, Verilog

Semiconductor & Verification Tools: Synopsys VCS, Verdi, Linux, Perforce, RTL automation, regression testing

Software & Data Systems: FastAPI, Streamlit, SQL, Git, data processing pipelines, statistical forecasting

Engineering Tools: MATLAB Simulink, LabVIEW, ANSYS, Fusion 360

Work Experience

Assistant Project Engineer, MSIP Department Jul 2024 – Present
Advanced Micro Devices, Inc. (AMD) Shanghai / London

- Contributed to memory read/write IP development for AMD accelerator projects.
- Developed automation and AI-assisted verification scripts for large-scale SoC signal inspection and traffic analysis.
- Built automated RTL error-injection frameworks to improve verification robustness and functional coverage evaluation.
- Maintained regression and validation workflows using Linux, Shell scripting, Python, Perforce, Synopsys VCS, and Verdi.
- Improved internal AI scripting tools by integrating intelligent command correction features to reduce user input errors.

Projects

Full-Stack Immigration Data Management System | FastAPI / Streamlit / SQL Sept 2025 – Apr 2026

- Developed a full-stack data management platform for the Singapore Immigration & Checkpoints Authority (ICA).
- Designed backend APIs and database structures using FastAPI for traveller record management and analytics.
- Built Streamlit dashboards for operational monitoring, data interaction, and visual analysis.
- Implemented short-term forecasting models to estimate future inbound traveller volume trends.
- Integrated automated data-processing pipelines to support decision-making workflows.

Energy-Aware Route Optimisation for Electric Scooters | Graph Algorithms / OSM Oct 2025 – Jun 2026

- Developed a simulation framework for electric scooter energy modelling using real-world map and elevation data.
- Implemented graph-based routing algorithms with custom energy-consumption edge weights derived from terrain and traffic conditions.
- Designed a hybrid optimisation strategy combining regenerative braking analysis with energy-aware route planning.
- Built automated simulation pipelines for large-scale route evaluation and statistical comparison across urban scenarios.

Path Planning for Trailer Obstacle Avoidance | A* Algorithms / Kinematic Constraints Sept 2022 – Jun 2023

- Developed a constrained path-planning framework for articulated trailer systems using an enhanced A* search algorithm.
- Integrated non-holonomic vehicle dynamics, steering continuity constraints, and collision-envelope detection into the planning heuristic.
- Generated dynamically feasible obstacle-avoidance trajectories while balancing computational efficiency and path optimality.

Leadership & Activities

UCL Racing Formula E Team Nov 2023 – Jun 2026

- Assisted in debugging motor signals and sensor feedback systems for the Formula E race car.

Vice President & Director of Program Control, Robotics Club Sept 2020 – Jun 2023

- Led the construction and debugging of robotic control modules and competed in the FRC China Regional Competition.

Honors & Awards

British Physics Olympiad Gold Award Nov 2022

Additional Information

Languages: Mandarin Chinese (Native), English (IELTS 7.5)

Interests: Football, Table Tennis, Musicals, Travelling